B. Tech Degree IV Semester Examination April 2011

ME 403 APPLIED THERMODYNAMICS

(2002 Scheme)

Time : 3 Hours | Maximum Marks : 100

I. (a) Derive the general steady flow energy equation. (10)
(b) Define (i) Reversibility (ii) Availability (10)

OR

II. (a) Write notes on (i) thermodynamic temperature scale (ii) Clausius claypeyron equation. (10)
(b) Derive Tds equations. (10)

III. (a) State and explain (i) Daltons law of partial pressure (ii) Amagats law. (10)
(b) Volumetric analysis of a gas mixture is as given below:
$N_2 = 82\%,\ CO_2 = 12\%,\ O_2 = 4\%$ and $CO = 2\%$
Determine analysis on a mass basis, molecular weight and gas constant for the exhaust gases. (10)

OR

IV. A vessel of 0.2 m^3 capacity contains 2 kg of CO_2 and 1.5 kg of N_2 at 300K. Determine (i) Pressure inside the vessel (ii) Mole fraction of each constituent (iii) Gas constant and molecular weight of mixture. (20)

V. Define (i) HCV and LCV (ii) Excess air (iii) Adiabatic flame temperature (iv) Enthalpy of combustion. (20)

OR

VI. The percentage composition by mass of a solid fuel is given below:
$C = 90\%,\ H_2 = 3.5\%,\ O_2 = 3\%,\ N_2 = 1\%,\ S = 1\%$ and the remainder being ash
(i) Find the mass of air required per kg of fuel for complete combustion
(ii) If 50% excess air is supplied in actual combustion, determine the volumetric analysis of the product and also the mass of flue gas per kg of fuel. (20)

VII. Write notes on (i) Morse test (ii) Heat balance sheet (iii) Scavenging (iv) Valve timing diagram (20)

OR

VIII. The following data is given for a single cylinder 4-stroke oil engine:
Cylinder diameter = 18cm, stroke = 36cm, speed of engine = 286 rpm,
brake torance =375Nm, IMEP=7 bar, Fuel consumption = 3.88 lph,
Sp.gravity of fuel=0.8, CV of fuel = 445000 kJ/kg, A:F=25:1,
Ambient temp.= 21^0C, Cpg = 1.2, Exhaust gas temp=415^0C,
cooling water circulated = 4.2kg/min., Rise in temperature of cooling water=28.5^0C.
Find mechanical and indicated thermal efficiency and draw heat balance sheet on percentage basis. (20)

IX. (a) Explain a Wankel engine. (10)
(b) Write notes on (i) Super charging (ii) Governing. (10)

OR

X. Briefly discuss (i) Preignition (ii) Diesel knock (iii) Cetane number (iv) HUCR (20)

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